

Role of fermentation pH in enhancing hydrogen production from orange peel using crude enzyme catalysis

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ABSTRACT

Dark fermentation is a promising strategy for biohydrogen production, but its low yield limits practical application. Crude enzyme extracted from orange peel has been reported to enhance hydrogen production rather than promote substrate hydrolysis during dark fermentation. Accordingly, this batch study investigated the influence of five initial fermentation pH levels (ranging from 5.5 to 9.5) on hydrogen production from synthetic swine wastewater via orange peel crude enzyme (OPCE)-mediated dark fermentation. The highest hydrogen yield (280.56 ± 5.34 mL/g glucose-added) was achieved at an initial pH of 6.5, where substrate conversion was most efficient and competing non-hydrogen pathways were minimized. Sludge at this pH exhibited a more favourable extracellular polymeric substance composition and structure, contributing to enhanced hydrogen production. A diverse microbial community was observed, with *Firmicutes* as the dominant phylum and *Clostridium saccharoperbutylacetonicum*, *Clostridium* sp., and *Clostridium butyricum* as key species, whose relative abundance varied with pH. These results highlight the critical influence of initial fermentation pH on optimizing OPCE-mediated dark fermentation for improved hydrogen yield.

List of abbreviation

CCUS	Carbon capture, utilization, and storage	OTUs	Operational taxonomic units
COD	Chemical oxygen demand	PN/PS	Protein/polysaccharide
EPS	Extracellular polymeric substances	PVPP	Polyvinylpyrrolidone
FID	Flame ionization detector	R ²	Correlation coefficient
GC	Gas chromatography	SDS-PAGE	Sodium dodecyl sulfate–polyacrylamide gel electrophoresis
LAB	Lactic acid bacteria	S-EPS	Soluble EPS

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LB-EPS	Loosely bound EPS	TB-EPS	Tightly bound EPS
NADH	Reduced nicotinamide adenine dinucleotide	TS	Total solids
OPCE	Orange peel crude enzyme	VFAs	Volatile fatty acids

1. Introduction

Exploring sustainable energy alternatives has become increasingly critical due to the depletion of fossil fuel reserves. Hydrogen is

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recognized as a promising substitute owing to its high energy density and clean combustion product—water. However, the environmental impacts of hydrogen vary substantially depending on the production technology and feedstock utilized [1]. To delineate these differences, hydrogen is categorized by color codes, such as grey, blue, pink, turquoise, purple, aqua, white, and green according to its primary energy source and production technologies [2]. For instance, grey hydrogen is derived from fossil fuels (coal gasification or steam reforming) without carbon capture, utilization, and storage (CCUS), while it turns blue if the production process combined with CCUS. Green hydrogen is typically produced through water electrolysis powered by renewable energy sources.

However, biohydrogen, which is produced from renewable raw materials through photo synthesis or fermentation, can also be regarded as green hydrogen since it most closely aligns with global carbon neutrality objectives [3]. Biohydrogen production has emerged as a key pathway for green hydrogen generation, with dark fermentation of organic waste representing a widely studied approach. Notably, the rapid expansion of pig farming has resulted in substantial quantities of organic-rich swine wastewater, offering a promising substrate for dark fermentative hydrogen production [4]. Through dark fermentation, organic matter in swine wastewater can be converted into hydrogen, volatile fatty acids (VFAs), and other value-added products, simultaneously reducing greenhouse gas emissions and mitigating environmental impacts [5].

Swine wastewater has been extensively studied as a substrate for dark fermentation due to its high organic content. However, hydrogen yields through dark fermentation remain low, predominantly due to the prevalence of low-yield metabolic pathways (e.g., butyrate-type fermentation) and competition from alternative pathways such as lactic acid and propionate production [6]. This limitation hinders the scalability of biohydrogen production. Various strategies—including genetic engineering, optimization of fermentation conditions, and the use of additives—have been explored to address this challenge.

Biohydrogen production is catalyzed by various hydrogen-producing enzymes, whose activities are strongly influenced by environmental factors such as temperature and pH. Additional fermentation parameters, including substrate type, organic loading rate, and carbon-to-nitrogen ratio, also play critical roles in optimizing dark fermentation performance [7]. Additive supplementation has emerged as another effective approach for enhancing hydrogen production. Appropriately dosed heavy metals, nanoparticles, biochar, and nitrous acid have been shown to improve the activity of key enzymes, including ferredoxin oxidoreductase, hydrogenase, and aldehyde dehydrogenase, by facilitating electron transfer, reducing inhibitory byproducts, or stabilizing enzyme structures [6].

Beyond chemical and nanomaterial additives, direct enzyme supplementation has recently gained attention. Most studies primarily employ exogenous enzymes, such as cellulase for cellulose [8], protease for chitosan [9], and lipase for palm oil mill effluent [10], to accelerate the hydrolysis of complex organic substrates, thus increasing the availability of fermentable substrates and improving hydrogen yields. In contrast, studies investigating the influence of direct enzyme addition on the hydrogen formation during dark fermentation remain limited. Among the limited studies, a previous batch study reported that orange peel crude enzyme (OPCE) enhanced hydrogen production by facilitating hydrogen formation during dark fermentation, increasing hydrogen yield by $17.10 \pm 4.21\%$ from synthetic swine wastewater [11]. Given that pH plays a crucial role in shaping microbial structure, enzyme activity, and fermentation pathways, the influence of initial fermentation pH on OPCE-assisted dark fermentation hydrogen production remains largely unexplored. This critical knowledge gap restricts both mechanistic understanding and process optimization of enzyme-assisted hydrogen production. Elucidating this relationship is essential not only for optimizing fermentation conditions and improving system performance, but also for advancing the practical deployment of

enzyme-assisted biohydrogen production technologies.

Therefore, this study investigates the effects of initial fermentation pH on OPCE-mediated dark fermentation with the aim of optimizing hydrogen production. Specifically, hydrogen yield and substrate utilization were first monitored under different initial pH conditions. Then, the composition and concentration of metabolic products, including VFAs and alcohols, were analyzed to clarify the metabolic pathways involved. Chemical oxygen demand (COD) mass balance was calculated to assess the efficiency of substrate-to-hydrogen conversion. Furthermore, changes in sludge physicochemical properties were characterized by analyzing extracellular polymeric substances (EPS) and zeta potential. Finally, microbial community dynamics were examined to reveal the underlying microbial mechanisms influenced by initial pH. This comprehensive study provides insights into optimizing enzymatic strategies for enhanced biohydrogen production and contributes to advancing the sustainable utilization of organic waste.

2. Material and methods

2.1. Inoculum and synthetic swine wastewater

The inoculum was anaerobic sludge obtained from the Wollongong wastewater treatment plant in Wollongong, Australia. The inoculum was stored at 4 °C before being used. Before the experiment, the inoculum was heated at 90 °C for 20 min to inactivate methanogens [12]. Synthetic swine wastewater was employed as the substrate for hydrogen production, with glucose serving as the primary carbon source, consistent with a previous study [13]. Given the presence of trace heavy metals in the sludge, no additional metal ions were supplemented to prevent potential interference while preserving hydrogen production efficiency. The components are as follows: 6.36 g/L glucose (about 6000 mg/L COD), 446 mg/L NH_4Cl , 132 mg/L KH_2PO_4 , 54 mg/L $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ and 4 mg/L $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ [11].

2.2. Orange peel crude enzyme extraction

Crude enzyme is a mixture of enzymes extracted from fresh orange peel using a modified method based on Feng et al. (Fig. 1) [11]. Initially, the orange peel was thoroughly washed with deionized water and then dried using napkins. Then, 50 g of orange peel was cut into small pieces and mixed with 500 mL 4 °C 0.02 M KH_2PO_4 solution and 0.5 g of polyvinylpyrrolidone (PVPP). The mixture was homogenized using a blender. Subsequently, the homogenate was filtered through four layers of gauze and centrifuged at 3500 rpm for 15 min to collect the supernatant. Next, ammonium sulfate was added for protein precipitation, with the saturation range of ammonium sulfate being 100%. The precipitated proteins were gathered and resuspended in 4 °C 0.02 M KH_2PO_4 . The solution was then dialyzed using a dialysis bag with a molecular weight cut-off of 5000 at 4 °C overnight, with the dialysis buffers 0.02 M KH_2PO_4 being changed every three to 4 h. The resulting dialysate was collected and centrifuged at 3500 rpm for 5 min. The supernatant was collected as crude enzyme and stored at 4 °C for future use. Sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS-PAGE) was employed to analyze the enzyme composition of OPCE.

2.3. Experimental set-up

500 mL serum bottles wrapped with black tape were used as dark fermentation bioreactors (Fig. 1). The working volume was 300 mL. OPCE was added to the bioreactors, and its dosage was standardized based on the protein concentration of OPCE, which was determined prior to the experiments as described in Section 2.4. The final volatile solids and protein concentrations in the bioreactors were about 1 g/L and 30 mg/L, respectively. The initial fermentation pH was adjusted using 0.1 M HCl and NaOH solution to 5.5, 6.5, 7.5, 8.5, and 9.5, respectively. After that, nitrogen gas was purged into the bioreactors for

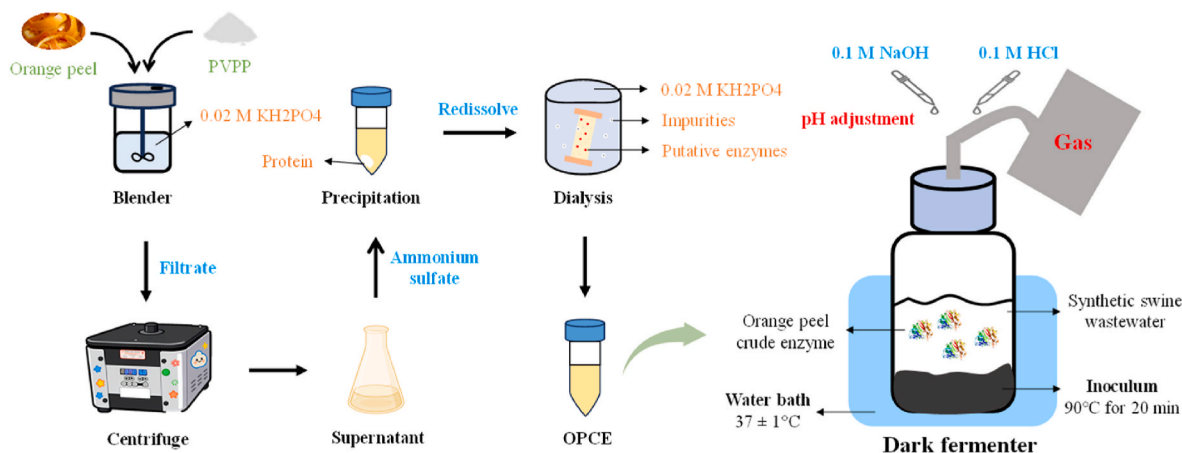


Fig. 1. Schematic diagram of the OPCE extraction procedure (left) and the dark fermenter used in this study (right). The dialysis used a 5 kDa cutoff membrane, retaining macromolecules, including enzymatic components, while smaller impurities were removed. PVPP: polyvinylpyrrolidone.

5 min at a flow rate of 25 mL/min to create an anaerobic condition. The experiment was conducted at 37 ± 1 °C using a water bath for 120 h without agitation in triplicate. Biogas was collected twice per 24-h cycle: first at the 6-h mark, and then again at the 18-h mark.

2.4. Analytical methods

The pH was detected using a pH meter (PHB-1, SUNNE). The total solids (TS) of the dark fermenter were measured according to the following processes. The EPS extraction was modified based on a previous study [14]. Briefly, 50 mL of sludge sample was collected at 54 h and then centrifuged at 4000 g for 5 min, and the supernatant was collected as soluble EPS (S-EPS). The sludge pellet left in the centrifuge tube was mixed with 50 mL of 70 °C NaCl (0.05% wt) solution. Then, the sludge suspension was vortexed and incubated for 1 min, followed by centrifugation at 4000g for 10 min. The supernatant was collected as the loosely bound EPS (LB-EPS). The sludge pellet left was re-suspended to its original volume by adding NaCl (0.05% wt) solution and placed in a water bath at 60 °C for 30 min, followed by centrifugation at 4000g for 15 min. The supernatant was collected as the tightly bound EPS (TB-EPS). All the supernatants were filtered by using 0.22 µm filters before analysis.

Residual glucose in the dark fermenters, and polysaccharides present in crude enzymes and EPS, were quantified using the modified phenol-sulfuric acid method, with glucose as the standard solution [15]. Specifically, 2 mL of samples was mixed with 0.8 mL of 5w% aqueous solution of redistilled phenol in a test tube. Then, 5 mL of concentrated sulfuric acid was added to the test tube with the mixture. After that, the test tube was placed in a boiling water bath for 20 min. Next, the test tube was cooled in an ice bath for 2 min and placed at room temperature for 15 min. Finally, the light absorption at 490 nm was measured on the spectrophotometer, and the carbohydrate concentration was determined according to a standard curve.

The BCA protein assay kit (KeyGEN BioTECH) was used to detect the concentration of proteins in crude enzymes and EPS. The zeta potential of the inoculum was detected using the zeta sizer (Malvern Zetasizer Nano). The volume of produced biogas was detected using a syringe, while the gas composition was measured by gas chromatography (GC) (Nexis GC-2030, SHIMADZU). The volume of produced hydrogen was calculated by multiplying the measured biogas volume by the detected hydrogen concentration. Collected samples were centrifuged at 10000 rpm for 5 min to remove the solid particles, and then the supernatant was filtered using 0.22 µm filters to remove the solid particles further. The filtered liquid samples detected glucose content, COD, and metabolites concentration. The glucose concentration was monitored following the phenol-sulfuric acid method with glucose as a standard

solution. COD was determined using HACH methods and test kits (200–15,000 mg/L) with a DR 3900 spectrophotometer (Hach). Metabolite concentrations, including VFAs and alcohols, were detected on a GC-MS (GC-2010 Plus, SHIMADZU, Japan) equipped with a flame ionization detector (FID) and a capillary column (SH-Stabliwax-DA, 30 m × 0.25 µm).

2.5. COD balance calculation and kinetics of hydrogen production

The calculation of COD balance follows by equation (1) [16]:

$$COD_{balance} = \frac{COD_{H_2} + COD_{final}}{COD_{initial}} \times 100\% \quad (1)$$

where COD_{final} and $COD_{initial}$ mean the COD in the effluent and influent, respectively. COD_{H_2} represents the COD converted to hydrogen. The specific conversion ratio of hydrogen is 8.0 g COD/g H_2 . The H_2 gas density is 0.08 g/L at 37 °C.

Biohydrogen production was fitted according to the modified Gompertz equation (2) [17] using Origin 2021 to acquire the kinetic parameter for biohydrogen production.

$$H_t = H_m \times \exp \left\{ - \exp \left[\frac{e \times R_m}{H_m} (\lambda - t) + 1 \right] \right\} \quad (2)$$

where H_t means the cumulative hydrogen yield (mL/g glucose-added), H_m is the highest hydrogen yield (mL/g glucose-added), R_m represents the maximum hydrogen yield rate (mL/g glucose/h), e is 2.718, λ and t are the hydrogen production lag time (h) and fermentation time (h), respectively.

2.6. Microbial community analysis

Sludge samples were collected at 54 h to assess the effects of fermentation pH on the microbial community, when biogas production had nearly ceased. The collected samples were centrifuged at 3000 rpm for 5 min, and the supernatant was discarded. The remaining solids were stored at -80 °C for further analysis. According to the manufacturer's instructions, DNA was extracted from the sludge samples using the E.Z. N.A. Soil DNA Extraction Kit (Omega Biotek, Norcross, GA, U.S.). PCR amplification of the V3-V4 region of the 16S rRNA gene was carried out using primers 341F (5'-CCTACGGGNGGCWGCAG-3') and 805R (5'-GACTACHVGGGTATCTAATCC-3'). The resulting PCR products were purified with Agencourt AMPure XP beads and subsequently used for library preparation. Library quality was assessed using the Agilent 2100 Bioanalyzer, and sequencing was performed on the DNBSEQ platform with paired-end reads.

Raw sequencing data were processed to remove low-quality reads (defined as an average quality score <20 over a 25 bp sliding window), adapter contamination, ambiguous bases, and low-complexity sequences using iTools Fqtools fqcheck (v0.25), cutadapt (v2.6), and readfq (v1.0). Paired-end reads were merged using FLASH (v1.2.11) with a minimum overlap of 15 bp and a maximum mismatch rate of 10%. High-quality merged sequences were clustered into operational taxonomic units (OTUs) at 97% similarity threshold using USEARCH (v7.0.1090). Chimeric sequences were identified and removed with UCHIME (v4.2.40). Representative sequences from each OTU were taxonomically classified using the RDP Classifier (v2.2) against the SILVA_16S database. OTUs not assigned to the bacterial domain were excluded from further analysis. Alpha diversity indices and taxonomic composition were analyzed based on the OTU abundance table and annotation results.

2.7. Calculations and statistical analysis

Figures in this study were made by the software Origin 2021. Statistical analysis of the data was performed using IBM SPSS Statistics version 28.0.0.0 (190). For datasets obtained from biological triplicates ($n = 3$), results are expressed as mean values with corresponding standard deviations. Differences among groups were evaluated using one-way analysis of variance (ANOVA), followed by Bonferroni post hoc tests. Statistical significance was determined at a confidence level of $p \leq 0.05$.

3. Results and discussion

3.1. SDS-PAGE analysis

To verify the presence of enzymes in the OPCE, SDS-PAGE analysis was performed. As shown in Fig. S1, a dominant protein band with an approximate molecular weight of 26 kDa was observed, consistent with the reported size of citrus peroxidase [18]. This result confirms the presence of protein components in OPCE, supporting its role as a bioactive additive in subsequent fermentation experiments.

3.2. Biohydrogen production

Since the inoculum was thermally pretreated, no methane was detected in the produced gas. The initial fermentation pH had a significant impact on hydrogen yield. At initial pH values of 5.5, 6.5, 7.5, 8.5, and 9.5, the corresponding hydrogen yields were 98.51 ± 1.41 , 280.56 ± 5.34 , 254.74 ± 10.50 , 136.11 ± 9.69 , and 238.01 ± 3.05 mL/g glucose-added, respectively (Fig. 2a). In our previous study, dark fermentation mediated by OPCE using synthetic swine wastewater achieved a hydrogen yield of 1.90 ± 0.08 mol H_2 /mol glucose (equivalent to 236.27 ± 9.95 mL H_2 /g glucose-added) at an initial pH of 6.0 [11]. The higher hydrogen yield observed at an initial pH of 6.5 in the present study underscores the importance of pH optimization in improving hydrogen production efficiency. The modified Gompertz equation was applied to further analyze the impact of initial pH on hydrogen production dynamics, achieving a correlation coefficient (R^2) more than 0.9804 (Table S1). The calculated results closely matched the

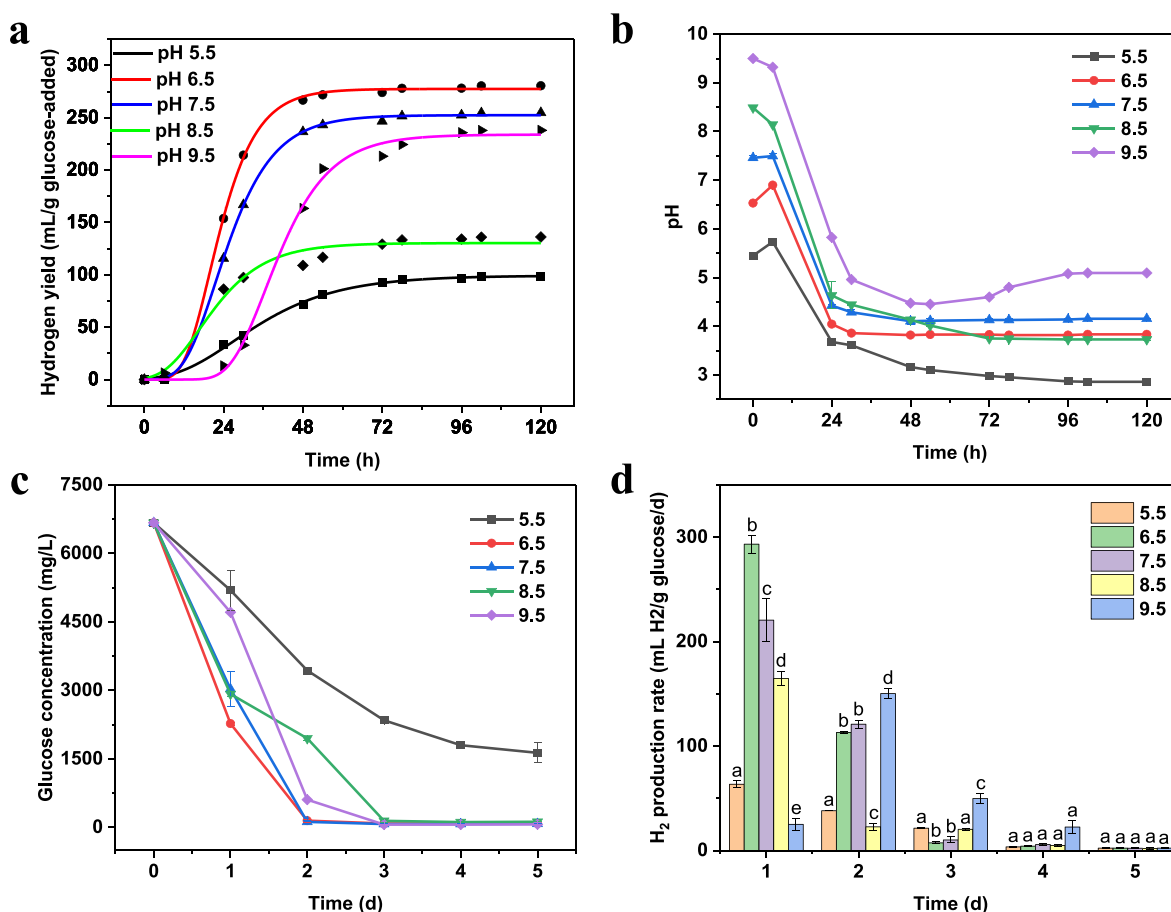


Fig. 2. Variations in hydrogen yield (a), pH (b), glucose concentration (c), and hydrogen production rate (d) during dark fermentation of synthetic swine wastewater under different initial fermentation pH conditions. The hydrogen production curves were fitted by the modified Gompertz equation. In Fig. 2b, error bars are present but not clearly visible due to their small magnitude. Different letters (a–e) in Fig. 2d indicate significant differences between groups ($\alpha = 0.05$).

maximum hydrogen yield observed in this research, confirming the data reliability. These findings indicate that an initial pH of 6.5 was optimal for hydrogen production, while an initial pH of 5.5 was not conducive to efficient hydrogen generation. This highlights the critical role of pH in optimizing fermentation conditions for biohydrogen production.

Hydrogen production primarily occurred within the first 72 h, with significant differences observed among the five pH groups. An initial fermentation pH of 6.5 achieved the highest hydrogen production rate of 293.36 ± 8.78 mL H₂/g glucose-added/d during the first 24 h of fermentation and maintained a peak rate of 13.11 ± 0.79 mL/g glucose-added/h (Table S1). Despite a lag time of approximately 12 h, this group achieved the shortest fermentation period of about 48 h (Fig. 2a), confirming an initial pH of 6.5 was optimal for hydrogen production, consistent with a previous study [19]. The hydrogen production rate at an initial pH of 7.5 was slightly lower than 6.5 during the first day. This was supported by the highest glucose consumption observed under an initial pH of 6.5 (Fig. 2c and d). However, on the second day, the fermenter at pH 7.5 consumed more glucose (approximately 780 mg/L) than that at pH 6.5 ($p < 0.05$), while exhibiting a similar hydrogen production rate (Fig. 2c and d). This suggests competing metabolic pathways for hydrogen production under an initial pH of 7.5.

In this study, an initial fermentation pH of 5.5 was not suitable for dark hydrogen fermentation. During the first day, the hydrogen production rate was 63.48 ± 3.49 mL H₂/g glucose/d, gradually decreasing to 2.30 ± 0.51 mL H₂/g glucose-added/d with the proceed of fermentation (Fig. 2d). An initial pH of 5.5 was reported to fall within the optimal pH range for dark fermentation [6]. However, the pH was decreased to below 3.68 after 24 h (Fig. 2b). The low pH penetrated the

microbial cell membrane and disrupt normal metabolic activities [20], resulting in low hydrogen production and a high amount of glucose in the effluent (Fig. 2c).

Unlike other pH conditions, the pH 8.5 group showed a hydrogen yield that deviated from the overall trend. The hydrogen production rate at pH 8.5 was 164.79 ± 6.43 mL H₂/g glucose-added/d on the first day, but it decreased to about 20 mL H₂/g glucose-added/d on the second and third days (Fig. 2d). The dark fermenter showed the shortest lag time of 5.91 ± 3.24 h, suggesting that this pH was suitable for the growth of most microorganisms. Moreover, it consumed a similar amount of glucose compared to the pH 7.5 group but with a lower hydrogen yield (Fig. 2a and c), indicating the presence of non-hydrogen-producing microorganisms. Notably, the pH dropped to about 4.6 at 24 h and decreased to around 4.1 at 48 h (Fig. 2b). This sharp drop in pH and significant lower hydrogen yield suggests the accumulation of acidic metabolites unrelated to hydrogen production, such as propionate and lactate [6], resulting in a lower hydrogen yield compared to the groups with initial pH values of 6.5, 7.5, and 9.5.

A lower glucose consumption was observed at an initial fermentation pH of 9.5 compared to the groups with initial pH values of 6.5, 7.5, and 8.5 during the first 24 h of fermentation (Fig. 2c). This is because a pH of 9.5 is typically not conducive to dark fermentation and inoculum growth [6], resulting in the most extended lag time (26.05 ± 1.14 h) (Table S1) and the lowest hydrogen production rate (25.04 ± 5.99 mL H₂/g glucose-added/d) during the first day (Fig. 2d). Despite the low glucose consumption and hydrogen yield observed at an initial fermentation pH of 9.5 during the first day, this group still produced acidic metabolites, such as acetate and butyrate, resulting in a pH drop to approximately

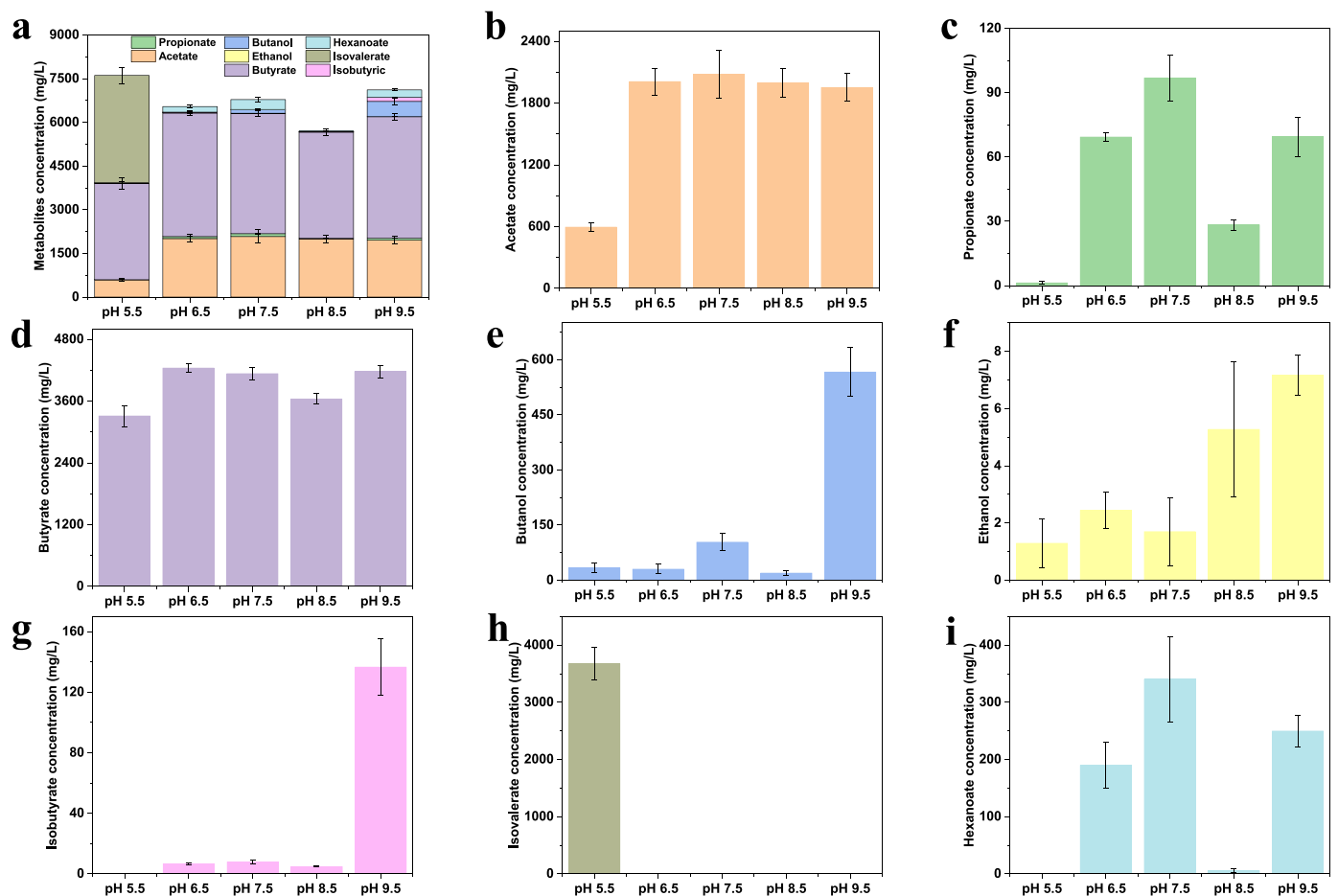


Fig. 3. Concentrations of total metabolites (a), acetate (b), propionate (c), butyrate (d), butanol (e), ethanol (f), isobutyrate (g), isovalerate (h), and hexanoate (i) after 120 h of fermentation under different initial pH conditions.

5.8 at 24 h (Fig. 2c), a range favourable for dark fermentation [6]. Consequently, a sharp decline in glucose concentration and a substantial increase in the hydrogen production rate were observed during the second day.

3.3. Metabolites production from glucose

After fermentation, the composition and concentration of metabolites in the effluent—significantly influenced by the initial fermentation pH—were analyzed (Fig. 3). Acetate and butyrate, which are typically associated with hydrogen generation during dark fermentation, were the predominant metabolites across all groups (Fig. 3). This observation aligns with previous studies [21,22], which have consistently reported acetate and butyrate as the major products of hydrogen-producing fermentation. Trace amounts of propionate, isobutyrate, and ethanol were also detected in all groups but had minimal impact on hydrogen yield (Fig. 3). Notably, hexanoate production, ranging from 190 to 340 mg/L, was observed at initial fermentation pH values of 6.5, 7.5, and 9.5 (Fig. 3), suggesting that an elongation of acetate likely occurred [23].

At an initial pH of 5.5, the main metabolites were acetate (594.32 ± 41.55 mg/L), butyrate (3311.44 ± 205.27 mg/L), and isovalerate (3675.52 ± 276.98 mg/L) (Fig. 3). Compared to the other groups, acetate and butyrate concentrations were significantly lower ($p < 0.05$), correlating with reduced hydrogen yields. Interestingly, a substantial accumulation of isovalerate occurred exclusively at pH 5.5, whereas its concentration remained below 6 mg/L in all other conditions (Fig. 3). Although *Clostridium saccharoperbutylacetonicum*, a known isovalerate-producing strain [24], was present across all groups (Section 3.6), the pronounced accumulation in this group appears to be primarily driven by the persistently low pH throughout fermentation and rapid acidification in the first two days (Fig. 2b). Under such acidic conditions, conventional acetate- and butyrate-type pathways were suppressed. As glucose continued to be consumed during the whole fermentation (Fig. 2c), metabolic flux was likely redirected toward branched-chain fatty acids, such as isovalerate. This interpretation is based on the observed distribution of fermentation products and established dark fermentation metabolic frameworks. However, direct analysis of metabolic pathways was not conducted in this study, and further investigations integrating functional or pathway-level analyses are required to confirm the proposed mechanisms.

In contrast, at initial pH values of 6.5, 7.5, and 9.5, acetate and butyrate were produced at comparable levels (~ 2000 mg/L and ~ 4200 mg/L, respectively) ($p > 0.05$) (Fig. 3), indicating similar fermentation profiles. However, hydrogen yields were significantly higher at pH 6.5 than that at pH 7.5 and 9.5 (Fig. 2a). Notably, acetate can also be produced by homoacetogens through non-hydrogen-producing pathways, with their optimal growth pH typically ranging from 5.5 to 7.0 [25,26]. Given that the genus *Clostridium*, which was detected in the inoculum (Section 3.6), includes known homoacetogenic species [27], the higher initial pH in the 7.5 and 9.5 groups may have favored homoacetogenic activity during the early fermentation stage, thereby contributing to reduced net hydrogen production. Moreover, the highest butanol concentration (518.93 ± 133.93 mg/L) was observed at pH 9.5, potentially contributing to lower hydrogen yield by consuming reduced nicotinamide adenine dinucleotide (NADH), a critical cofactor in hydrogen production [28]. Thus, competing metabolic pathways likely played a role in the reduced hydrogen yields at pH 7.5 and 9.5 compared to pH 6.5.

For the group with an initial pH of 8.5, although acetate production was comparable to other groups, butyrate concentration was significantly lower (3642.20 ± 103.77 mg/L) (Fig. 3). This suppression of butyrate-type fermentation likely contributed to the group's low hydrogen production. Furthermore, despite complete glucose consumption, the total concentration of detected metabolites was markedly lower than in the other four groups, suggesting the formation of undetected by-products. Compared to the pH 7.5 group, the pH 8.5 group

showed similar glucose consumption, but lower hydrogen yields and a more rapid pH decline on the first day (Fig. 2), indicating the possible accumulation of non-hydrogen-associated acidic metabolites, such as lactate. Although acidic conditions (pH 5.5–6.0) generally favor lactic acid bacteria (LAB) [29], some LAB strains can tolerate alkaline environments up to pH 8.9 [30]. Therefore, lactate production could have occurred at pH 8.5, leading to pH decline, altered fermentation pathways, and suppressed hydrogen production. Several LAB were detected in this group (Section 3.6). Moreover, other acidic metabolites, such as succinate, also compete with hydrogen production. However, the concentrations of these metabolites were not measured, representing a limitation of this study that should be addressed in future research.

3.4. Effects of initial fermentation pH on COD balance

Initial fermentation pH significantly influenced substrate utilization. As shown in Table 2, all five groups exhibited a COD balance of over 86%, indicating reliable data and measurements for dark hydrogen production [31]. Among the tested conditions, an initial fermentation pH of 6.5 resulted in the highest COD-to-hydrogen conversion efficiency ($25.33 \pm 1.14\%$), along with the lowest residual COD in the effluent (Table 1). Although the group with an initial pH of 8.5 achieved a slightly higher conversion efficiency ($12.27 \pm 0.55\%$) than the pH 5.5 group, its overall performance remained inferior to that observed at pH 6.5, 7.5, and 9.5 (Table 1). Notably, the pH 8.5 group experienced the most pronounced pH decline, dropping sharply from 8.5 to approximately 4.6 within the first 24 h (Fig. 2b). This rapid and substantial decrease likely stressed the microbial community, disrupted intracellular homeostasis, impairing growth, and potentially redirecting metabolic pathways, consistent with previous reports on microbial responses to pH shock [32]. These findings underscore that both the magnitude and rate of pH change during fermentation are critical determinants of hydrogen production efficiency.

3.5. EPS and zeta potential

The initial fermentation pH significantly influenced the composition of the sludge EPS. EPS, composed of complex extracellular polymers, form a protective matrix that shield cells from abiotic stress and supports cell survival and growth, particularly under adverse conditions [33]. As shown in Table S2, S-EPS was the primary component of total EPS in all groups, whereas LB-EPS and TB-EPS contributed only a minor proportion. Therefore, total EPS was used to describe the overall EPS response under different initial pH conditions. In this study, the sludge at an initial fermentation pH of 5.5 exhibited the highest EPS concentration, coinciding with a drop in fermentation pH to approximately 3 by the sampling time (Fig. 2b). The elevated EPS levels under such acidic conditions are likely a stress response by the hydrogen-producing microbial consortium, aiming to protect cells from the harsh environment. The second-highest EPS concentration was observed at an initial pH of 8.5, implying that the drastic pH decline (Fig. 2b) similarly stressed the microbial community and altered their metabolic activity, consistent with previous findings on microbial responses to pH shock [32]. For initial fermentation pH values of 6.5, 7.5, and 9.5, the total EPS concentration decreased with increasing initial pH (Fig. 4a), likely because higher initial pH levels led to higher fermentation pH levels at the sampling time (Fig. 2b).

Polysaccharides in EPS provide a robust and adhesive framework crucial for granular sludge formation, while proteins are more strongly associated with the formation and stability of microbial aggregates [34]. High EPS protein content has been reported to enhance microbial aggregation and matrix integrity [35]. In this study, groups with initial pH values of 6.5, 7.5, 8.5, and 9.5 displayed relatively high protein concentrations, approximately 425 mg/L for pH 6.5 and 8.5, and about 330 mg/L for pH 7.5 and 9.5 (Fig. 4a). Although microbial aggregation was not directly assessed, the higher protein content at pH 6.5 and 8.5 may

Table 1

COD mass balance for hydrogen production through dark fermentation of swine wastewater under different initial fermentation pH conditions.

	pH 5.5	pH 6.5	pH 7.5	pH 8.5	pH 9.5
Initial COD (mg)	2115.00 ± 55.15	2115.00 ± 55.15	2115.00 ± 55.15	2115.00 ± 55.15	2115.00 ± 55.15
COD _{final} (%)	77.31 ± 0.31 ^a	68.22 ± 0.23 ^b	69.67 ± 2.02 ^b	76.04 ± 0.78 ^a	75.04 ± 0.35 ^a
COD removal (%)	22.72 ± 2.68 ^a	31.81 ± 3.15 ^b	30.36 ± 2.18 ^b	24.00 ± 2.60 ^a	24.99 ± 2.82 ^a
H ₂ as COD (%)	8.89 ± 0.36 ^a	25.33 ± 1.14 ^b	22.98 ± 0.35 ^c	12.27 ± 0.55 ^d	21.48 ± 0.29 ^c
COD balance (%)	86.20 ± 0.97 ^a	93.55 ± 0.91 ^b	92.65 ± 1.67 ^b	88.31 ± 0.23 ^a	96.52 ± 0.64 ^c

Note: The symbols a, b, and c represent the differences between groups, with a significance level (α) of 0.05.**Table 2**

Species observed (sobs) and calculated richness and diversity estimators were obtained from the five samples at 54 h based on a threshold of 97% on the 16S rRNA gene level.

Sample Name	Sobs	Chao 1	ACE	Shannon	Simpson	Good's coverage
pH 5.5	207	325.32	438.37	1.35	0.34	0.998
pH 6.5	264	451.50	433.31	1.42	0.36	0.998
pH 7.5	258	367.60	369.50	1.59	0.32	0.998
pH 8.5	283	386.36	418.91	1.00	0.62	0.998
pH 9.5	370	448.92	454.49	2.35	0.17	0.998

be associated with enhanced stability of microbial aggregates. In contrast, the group at pH 5.5 showed markedly lower protein content and a higher polysaccharide proportion, possibly due to inhibited microbial growth and the accumulation of approximately 2300 mg/L of unutilized glucose (Fig. 2c), suggesting limited microbial activity and weaker EPS-mediated interactions under acidic conditions.

The protein/polysaccharide (PN/PS) ratio in EPS serves as an important indicator of sludge settling properties. A lower PN/PS ratio is generally associated with greater granule stability, whereas a higher ratio indicates reduced structural strength and poorer settling performance [36]. In this study, the PN/PS ratio decreased from 3.92 ± 0.08 to 2.24 ± 0.04 as the initial fermentation pH increased from 6.5 to 8.5 (Fig. 4a). A similar trend was reported by Sriwiriya and Nuchlek [37], who observed a decline in the PN/PS ratio from 0.12 to 0.08 (w/w) with increasing influent pH. However, an excessively high PN/PS ratio can impair sludge settling due to excessive protein content [34]. In this study, relatively poorer sludge settling at pH 6.5 and 9.5 compared to pH 7.5 may have resulted in a looser sludge structure, potentially improving substrate-microbe contact efficiency and contributing to the observed enhancement in short-term hydrogen production. It should be noted that

EPS in this study was characterized only by bulk measurements, and more detailed analyses such as three-dimensional fluorescence spectroscopy would be needed to further clarify its structural and functional features.

Zeta potential reflects the electrostatic stability of the sludge colloid matrix by indicating the surface charge of EPS, which affects particle aggregation and settling behavior [38]. A higher absolute zeta potential value implies stronger electrostatic repulsion between sludge flocs, leading to poor flocculation and reduced dewatering efficiency [39]. In this study, the absolute zeta potential generally increased with the initial fermentation pH (Fig. 4b). Specifically, the relatively low absolute zeta potentials at initial pH 5.5 and 8.5 suggest easier aggregation of sludge particles under these conditions. Although aggregation was not directly measured, reduced effective surface area for substrate-microbe interaction under these conditions could partly explain the lower hydrogen yields observed. The zeta potential values at initial pH 6.5 and 7.5 were similar (approximately -16.5 mV). Given that initial pH 6.5 yielded the highest hydrogen production, this suggests favourable microbial contact efficiency with the substrate. However, at an initial pH of 9.5, the zeta potential increased to -23.27 ± 2.60 mV. Although the stronger surface charge may have improved sludge-substrate contact, it could have weakened microbial synergistic interactions, leading to lower hydrogen production compared to the group at pH 6.5 [39].

3.6. Effects of initial fermentation pH on microbial community

Sludge samples were collected at 54 h to assess the microbial diversity and structure at five different initial fermentation pH levels (Table 2). The results showed that initial fermentation pH significantly influenced microbial diversity and biohydrogen production. In this study, Good's coverage for all samples was more significant than 0.998, suggesting these data's reliability in capturing the microbial community's diversity. At pH 5.5, the species richness (Sobs) and Chao 1 index

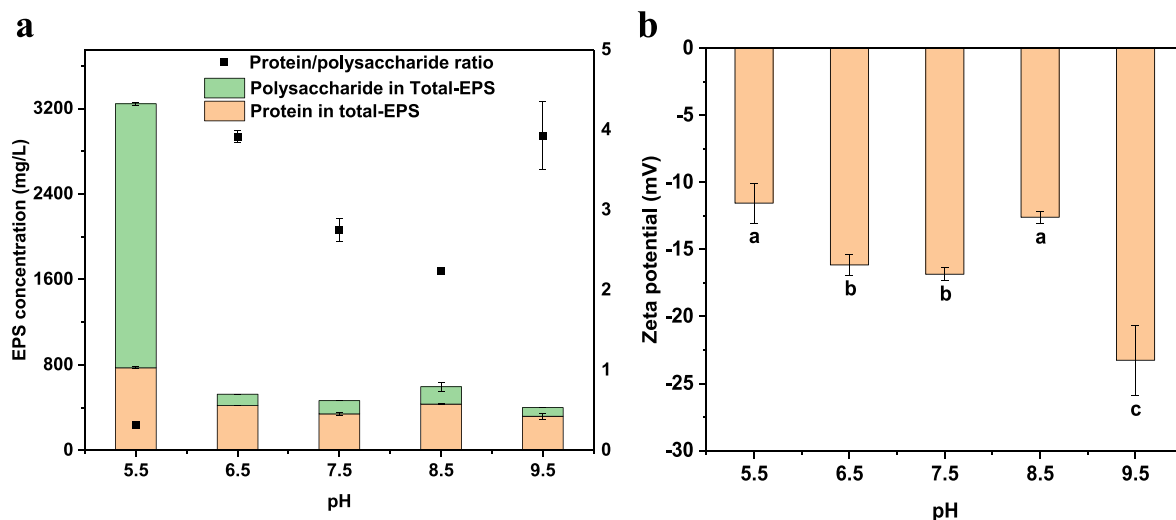


Fig. 4. EPS concentration (a) and zeta potential (b) of sludge in the dark fermenter under different initial fermentation pH conditions. The symbols a and b represent the differences between groups, with a significance level (α) of 0.05.

were the lowest (207 and 325.32, respectively), indicating that the acidic environment restricted microbial diversity. As pH increased, both Sobs and Chao 1 values rose, with the highest values recorded at pH 9.5 (370 and 448.92, respectively), suggesting that a more neutral to slightly alkaline pH fosters greater diversity. The Shannon index was highest at pH 9.5 (2.35), while the Simpson index was lowest (0.17), indicating a well-balanced microbial community. Despite the highest microbial diversity at pH 9.5, this condition did not favor hydrogen production. The high pH likely promoted non-hydrogen-producing metabolic pathways [40], as the optimal pH for dark fermentation typically lies between 5.0 and 6.5 [6]. In contrast, pH 6.5 showed the best hydrogen production, balancing microbial diversity and pathways favouring hydrogen production. This pH supported higher yields of biohydrogen, highlighting its optimal role in dark fermentation conditions.

The microbial community of all five groups at phylum, genus, and specie levels after 54 h of fermentation is shown in Fig. 5. At the phylum level, *Proteobacteria* and *Firmicutes* were dominant in all samples. *Proteobacteria* convert organic matter into VFAs and hydrogen during dark fermentation [41]. This phylum can form synergistic microbial communities with *Firmicutes*, enhancing the efficiency and stability of hydrogen production [42]. However, in the initial pH range of 5.5 to 8.5, the relative abundance of *Proteobacteria* was low, ranging from 0.58% to 2.23% (Fig. 5a). The highest relative abundance (8.08%) was observed at an initial fermentation pH of 9.5, suggesting that this condition favored the growth of *Proteobacteria*, thereby contributing to a more stable microbial community.

In this study, LAB such as *Olsenella* and *Bifidobacterium* were detected in the inoculum; however, their relative abundances remained below 0.27% across all groups (Fig. 5b). LAB generally exhibit optimal growth

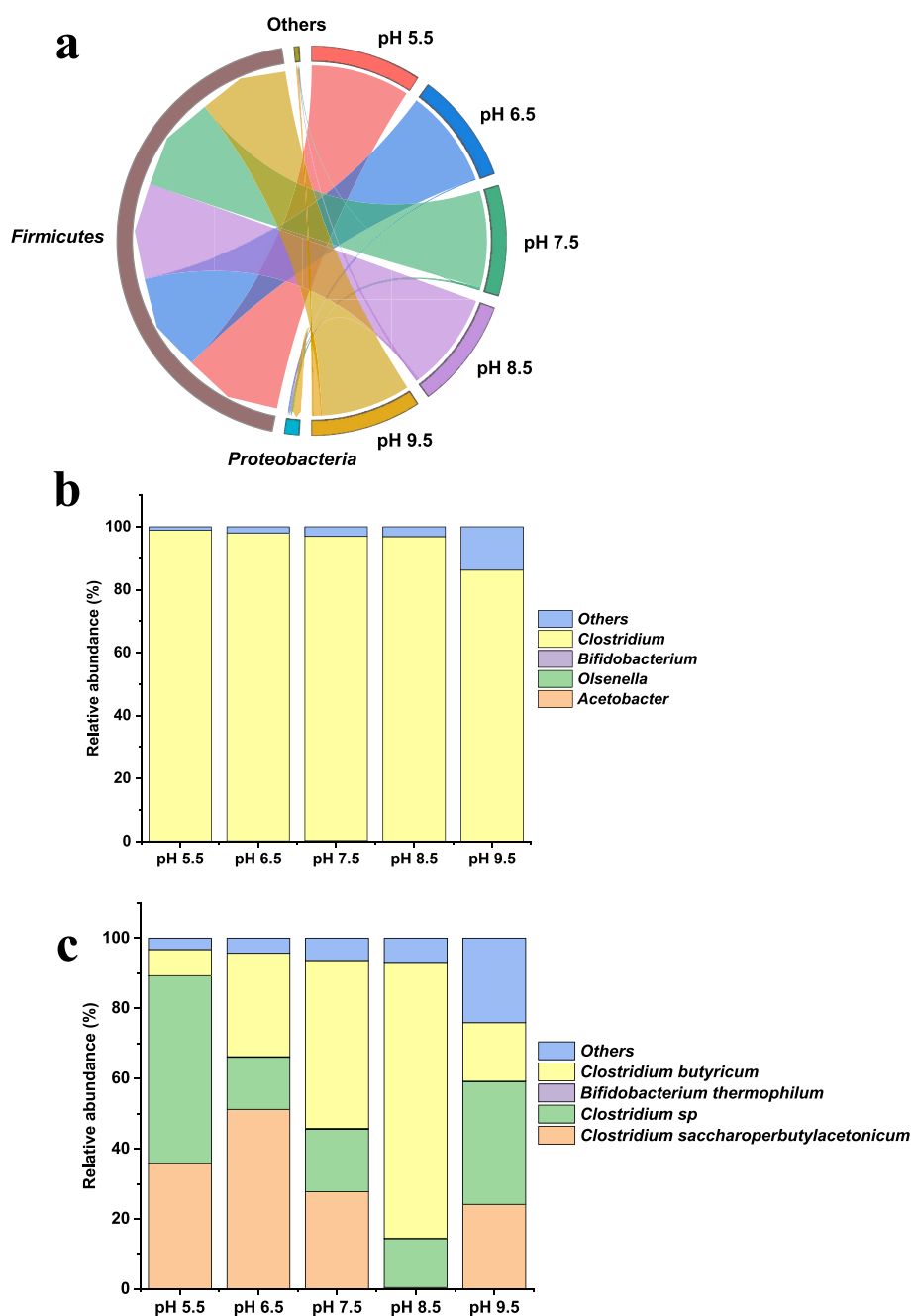


Fig. 5. Microbial community structure at the phylum (a), genus (b), and species (c) levels under different initial fermentation pH conditions.

at pH 5.5–6.0, and exposure to low pH can trigger acid stress responses, ultimately compromising their viability over prolonged periods [29]. As shown in Fig. 2b, the rapid pH decline driven by the accumulation of acidic metabolites created an unfavorable environment for LAB survival, explaining their consistently low abundance. Interestingly, certain LAB species are capable of growing under mildly alkaline conditions up to pH 8.9 [30], suggesting that the initial pH 8.5 condition may have supported lactate production. This observation aligns with our earlier inference that lactate accumulation was one of the contributors to the sharp pH drop in this group.

Firmicutes, a phylum known for its important role in dark fermentation hydrogen production, dominated the microbial community in all samples, consistent with findings from other dark fermenters [21,22]. The relative abundance of *Firmicutes* was approximately 97% across the initial fermentation pH ranged of 5.5 to 8.5 but decreased to 89.73% at pH 9.5 (Fig. 5a). Within this phylum, *Clostridium* was the predominant genus, with relative abundances of 98.78%, 97.84%, 96.64%, 96.73%, and 86.14% corresponding to initial pH values of 5.5, 6.5, 7.5, 8.5, and 9.5, respectively (Fig. 5b). The reduction in *Clostridium* abundance at pH 9.5 likely reflects the increased diversity and proliferation of other microbial groups under more alkaline conditions (Table 2). The consistently high relative abundance of *Firmicutes* and *Clostridium* across the studied pH range indicates that their growth was relatively resilient to pH variations from 5.5 to 9.5, highlighting their superior acid tolerance and metabolic versatility under anaerobic and acidic environments [43].

At the species level, the genus *Clostridium* was primarily composed of *Clostridium saccharoperbutylacetonicum*, *Clostridium* sp., and *Clostridium butyricum*, but their proportions varied significantly (Fig. 5c). The initial fermentation pH significantly affected the composition of microbial community, leading to a distinct biohydrogen production performance. Specifically, the relative abundance of *Clostridium saccharoperbutylacetonicum* at initial fermentation pH of 5.5, 6.5, 7.5, 8.5, and 9.5 was 35.86%, 51.22%, 27.74%, 0.44%, and 24.13%, respectively. A previous study reported that the maximum hydrogen yield and production rate using *Clostridium saccharoperbutylacetonicum* for hydrogen production from cheese whey occurred at an initial pH of 6, with a steady decline as the pH increased to 10 [44]. Consistent with these findings, this study observed that the relative abundance of *Clostridium saccharoperbutylacetonicum* peaked at an initial pH of 6.5, followed by a gradual decline as the pH increased to 9.5. Notably, at an initial fermentation pH of 8.5, the relative abundance of *Clostridium saccharoperbutylacetonicum* was drastically reduced (0.44%). This suppression may be attributed to environmental conditions favouring the preferential enrichment of *Clostridium butyricum*, which achieved its highest relative abundance (78.30%) at this pH. These results suggest that *Clostridium saccharoperbutylacetonicum* may be less competitive under alkaline conditions [44], potentially due to pH-induced shifts in substrate utilization or ecological niche competition. Further research is required to verify whether *Clostridium butyricum* directly outcompetes *Clostridium saccharoperbutylacetonicum* or whether additional factors contribute to this observed suppression.

The relative abundance of *Clostridium butyricum* increased markedly from 7.42% to 78.30% as the initial fermentation pH rose from 5.5 to 8.5 but then dropped sharply to 16.71% at pH 9.5 (Fig. 5c), indicating that a neutral to mildly alkaline environment favored its enrichment. Although *Clostridium butyricum* is recognized as an effective hydrogen producer [45], the group with an initial pH of 8.5 exhibited relatively low hydrogen yields. This outcome may be linked to a rapid decline in fermentation pH to around 4.6 within 24 h (Fig. 2b), likely driven by the accumulation of organic acids, particularly lactic acid, as evidenced by the presence of LAB (Fig. 5). While *Clostridium butyricum* can utilize lactic acid, its hydrogen production is primarily associated with carbohydrate metabolism, and it performs poorly in lactate-driven systems [46]. Thus, despite its high abundance at pH 8.5, the limited ability of *Clostridium butyricum* to convert lactic acid into hydrogen likely contributed to the observed low hydrogen yields.

Apart from the species *Clostridium saccharoperbutylacetonicum* and *Clostridium butyricum*, *Clostridium* sp. was also a significant component of the microbial communities. The highest relative abundance of *Clostridium* sp. was observed at an initial fermentation pH of 5.5 (53.40%), followed by pH 9.5 (35.06%). At initial fermentation pH levels of 6.5, 7.5, and 8.5, its relative abundance was similar, around 15% (Fig. 5c). *Clostridium* sp. is known for hydrogen production, typically associated with acetic acid as the primary metabolite [43]. This suggests that *Clostridium* sp. contributed to the acetate detected in this experiment, indicating its potential in hydrogen production through acetate-associated pathways. However, some strains within *Clostridium* sp. have been reported as homoacetogens under dark fermentation conditions, producing acetate at the expense of hydrogen [47]. This dual metabolic potential complicates the current study's interpretation of its role in hydrogen production. Further investigations are necessary to clarify the specific contributions of *Clostridium* sp. to hydrogen production.

4. Conclusion

This study demonstrates that the initial fermentation pH is a key operational parameter governing biohydrogen production from synthetic swine wastewater in an OPCE-mediated dark fermentation system. The main conclusions are summarized as follows:

Among the investigated pH range (5.5–9.5), the maximum hydrogen yield ($280.56 \pm 5.34 \text{ mL g}^{-1}$ glucose-added) was obtained at an initial pH of 6.5.

Operation at pH 6.5 enhanced substrate utilization efficiency and suppressed the formation of non-hydrogen by-products, particularly butanol, thereby favoring hydrogen-oriented metabolic pathways.

Improved EPS composition and a more favourable sludge structure at pH 6.5 likely strengthened microbial–substrate interactions, contributing to enhanced hydrogen production.

Higher microbial diversity and increased abundance of hydrogen-producing populations were observed at pH 6.5. Although *Firmicutes* remained the dominant phylum across all tested pH conditions, the relative abundance of key hydrogen-producing species, including *Clostridium saccharoperbutylacetonicum*, *Clostridium* sp., and *Clostridium butyricum*, was strongly influenced by pH.

Overall, these results provide important insights into the physico-chemical and microbial responses of OPCE-assisted dark fermentation systems to pH regulation. However, the mechanistic interpretations are mainly derived from process performance, metabolite distribution, and microbial community analyses. Future studies should integrate metabolic and functional approaches, particularly focusing on hydrogen-producing pathways and volatile fatty acid metabolism, to further elucidate pH-dependent regulatory mechanisms. Moreover, long-term stability and scalability of OPCE systems operated at optimal pH conditions should be systematically evaluated to support their application in sustainable biohydrogen production.

CRediT authorship contribution statement

Siran Feng: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Wenshan Guo:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation. **Huiying Zhang:** Writing – review & editing, Formal analysis, Data curation. **Guangming Zhang:** Writing – review & editing, Resources, Methodology. **Panyue Zhang:** Writing – review & editing, Methodology, Formal analysis. **Soon Woong Chang:** Writing – review & editing, Resources, Methodology. **Dinh Duc Nguyen:** Writing – review & editing, Formal analysis, Data curation. **Yuan Yao Ye:** Writing – review & editing, Resources, Methodology. **An Ding:** Writing – review & editing, Resources, Formal analysis. **Huu Hao Ngo:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.renene.2026.125685>.

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